Advanced Training for Oil Tanker Cargo Operations (TA

Advanced Training for Oil Tanker Cargo Operations (TASCO)

30 Practice Questions and Answers for STCW Exams

Updated for DG Shipping India Exams | *Source: brightmariner.com*

Practice Set 1

Q1. What is the main requirement of the OPA 90?

A) New tankers and tank barges should have double hulls
B) New tankers and tank barges should not ply in US waters
C) New tankers and tank barges with single hulls can ply in us ports

Correct Answer: A (New tankers and tank barges should have double hulls)

Explanation: The Oil Pollution Act of 1990 (OPA 90) was enacted in the United States following the Exxon Valdez incident. Its primary requirement is the mandatory fitting of double hulls on all new tankers and tank barges to prevent oil spills during groundings or collisions.

Q2. Volume 2 of IMDG code comprises one part which is

A) Part 1 - General provisions, definitions and training
B) Part 2 - Classification
C) Part 3 - The Dangerous Goods List (DGL)
D) Part 4 - Packing and tank provisions

Correct Answer: C (Part 3 - The Dangerous Goods List (DGL))

Explanation: Volume 2 of the IMDG Code is structured to provide technical guidance, specifically containing Part 3, which is the Dangerous Goods List (DGL). This list is essential for identifying the specific shipping requirements and classification of hazardous materials.

Q3. What is the maximum limit of sulphur content in bunker fuel oil?

A) $4.5\%m/m$
B) $1.5\%m/m$
C) $3.0\%m/m$

Correct Answer: A ($4.5\%m/m$)

Explanation: MARPOL Annex VI sets strict limits on the sulphur content of bunker fuel to reduce SOx emissions. The standard global limit, as established in the regulation, is 4.5% m/m, though regional Emission Control Areas (ECAs) now have significantly stricter requirements.

Q4. What is the minimum requirement for carrying garbage management plan onboard a ship?

A) Any ship above 400 GT or any ship carrying 15 persons or more
B) Any ship which is 12 metres or more in length
C) All ship, Irrespective of size and length

Correct Answer: C (All ship, Irrespective of size and length)

Explanation: According to MARPOL Annex V, all ships, regardless of size and length, are required to have a Garbage Management

Plan to ensure proper handling, separation, and disposal of waste materials to prevent marine pollution.

Q5. Petroleum liquids that have closed-cup flash points below 140 Deg F (60 Deg C) are considered as

A) Non-volatile Liquids
B) Flammable Liquids
C) Volatile Liquids
D) Combustible Liquids

Correct Answer: C (Volatile Liquids)

Explanation: Under standard classification, petroleum liquids with a closed-cup flash point below 60 degreesC are categorized as 'volatile' or 'flammable' because they can easily produce vapours that form ignitable mixtures with air at ambient temperatures.

Q6. What does SECA stand for?

A) Sox Elimination Controlled Area
B) Sox Emission Controlled Area
C) Sox Escalation Controlled Area

Correct Answer: B (Sox Emission Controlled Area)

Explanation: SECA stands for SOx Emission Control Area. Under MARPOL Annex VI, these are designated sea areas where stricter controls on sulfur emissions from ships are enforced to reduce air pollution and protect human health and the environment.

Q7. What are the three broad classification of cargoes transported on tankers?

A) Liquefied Natural gas, Liquefied Petroleum gas, Liquefied Nitrogen
B) Container, Dry Bulk and Refrigerated cargo
C) Petroleum liquids, chemical liquids, and special liquids
D) General cargo, Dry Break Bulk and Frozen cargo

Correct Answer: C (Petroleum liquids, chemical liquids, and special liquids)

Explanation: Tanker cargoes are broadly classified into petroleum liquids (crude oil and products), chemical liquids (noxious liquid substances), and special liquids (like vegetable/animal oils and molasses). This classification helps determine carriage requirements, hazard profiles, and equipment standards under IBC and IGC codes.

Q8. What is the closed-cup flash point of non-volatile petroleum liquids?

A) below 140 Deg F (60 Deg C)
B) 113 Deg F (45 Deg C) and above
C) below 113 Deg F (45 Deg C)
D) 140 Deg F (60 Deg C) and above

Correct Answer: D (140 Deg F (60 Deg C) and above)

Explanation: Non-volatile petroleum liquids are defined by their higher flash points. According to international safety standards for tanker operations, a liquid with a closed-cup flash point of 60 degreesC (140 degreesF) or above is considered non-volatile, reducing the risk of vapor accumulation at ambient temperatures.

Q9. What is the Reid vapor pressure of flammable liquids of natural gasoline & Naphtha which has open cup flash point of 80 Deg F (26.7 Deg C)?

A) More than 8.5 but less than 14 psi
B) 8.5 psi and below
C) 14 psi and above
D) More than 24.6 psi

Correct Answer: A (More than 8.5 but less than 14 psi)

Explanation: Reid Vapor Pressure (RVP) measures the volatility of petroleum products. For natural gasoline and Naphtha with a flash point of 80 degreesF, the specific regulatory classification for RVP range is categorized as more than 8.5 but less than 14 psi to determine handling safety.

Q10. What is the requirement for ships to have SOPEP onboard ships?

A) SOPEP Should be carried on board tankers of 150 GT or more and on other ships of 400 GT or more
B) SOPEP should be carried onboard tankers of 150 GT or more and on all other ships irrespective of size
C) SOPEP should be carried only onboard tankers of 400 GT or more

Correct Answer: A (SOPEP Should be carried on board tankers of 150 GT or more and on other ships of 400 GT or mo

Explanation: Under MARPOL Annex I, Regulation 37, every oil tanker of 150 GT or more and every ship of 400 GT or more is required to carry an approved Shipboard Oil Pollution Emergency Plan (SOPEP) to facilitate effective response to oil spills.

Q11. What is the meaning of a Category 2 ship?

A) Oil tanker of 5000 tones deadweight and above
B) Tanker of 20,000 deadweight and above without SBTs
C) Tanker of 20,000 deadweight and above with SBTs

Correct Answer: B (Tanker of 20,000 deadweight and above without SBTs)

Explanation: Under MARPOL Annex I, Category 2 ships refer to oil tankers of 20,000 tonnes deadweight and above that are not fitted with Segregated Ballast Tanks (SBT) in accordance with the regulation's phase-out requirements for single-hull tankers.

Q12. What type of ships have to carry the SMPEP for NLS manual onboard their ships?

A) Ship of 400 GT or more carrying 15 persons or more
B) Ships of 150 GT or more carrying oil cargoes
C) Ships of 150 Gt or more carrying noxious liquid substances

Correct Answer: C (Ships of 150 Gt or more carrying noxious liquid substances)

Explanation: Under MARPOL Annex II, ships of 150 GT or more that are certified to carry noxious liquid substances in bulk are required to have a Shipboard Marine Pollution Emergency Plan (SMPEP) for NLS on board.

Q13. Tankers carry palm oil, molasses and tallow as special liquids cargoes in bulk, which are grouped as

A) Animal/Vegetable oils
B) Miscellaneous Liquids
C) Non-volatile Liquids
D) Other chemical Liquids

Correct Answer: A (Animal/Vegetable oils)

Explanation: Palm oil, molasses, and tallow are classified as Animal and Vegetable oils. These are distinct from mineral oils and fall under specific requirements for carriage and pollution prevention on specialized tankers.

Q14. Under which Annex is the Garbage record book required?

A) Annex II
B) Annex V
C) Annex VI

Correct Answer: B (Annex V)

Explanation: MARPOL Annex V deals with the prevention of pollution by garbage from ships. This regulation mandates the maintenance of a Garbage Record Book to track the discharge or disposal of all garbage generated onboard.

Q15. Animal/Vegetable Oils carried in tankers and classified as

A) Chemical Liquid
B) Special Liquids
C) Non-volatile Liquids
D) Non-flammable Liquids

Correct Answer: B (Special Liquids)

Explanation: Animal and Vegetable oils, while distinct from petroleum hydrocarbons, are categorized as special liquids due to their persistent nature and specific requirements for carriage and potential pollution mitigation as defined in MARPOL and IBC Code guidelines.

Q16. Which part of the oil record book deals with machinery space operations?

A) Part I
B) Part II
C) Part III

Correct Answer: A (Part I)

Explanation: The Oil Record Book is divided into two parts: Part I (Machinery space operations) for all ships, and Part II (Cargo/ballast operations) for oil tankers. All entries regarding oil transfers, fuel bunkering, and bilge water discharge are strictly documented here.

Q17. Structure of MARPOL Convention layouts are as follows: find the correct one.

A) Convention, Protocol I, Protocol II, Protocol 1978, Protocol 1997
B) Convention, Protocol 1978, Protocol 1997, Protocol I, Protocol II
C) Protocol 1978, Protocol 1997, Protocol I, Protocol II, Convention
D) Protocol 1997, Protocol 1978, Protocol I, Protocol II, Convention

Correct Answer: B (Convention, Protocol 1978, Protocol 1997, Protocol I, Protocol II)

Explanation: The formal structure of the MARPOL Convention includes the main Convention, Protocol I (Reports on Incidents), Protocol II (Arbitration), and subsequent major amendments including the 1978 Protocol (MARPOL 73/78) and the 1997 Protocol (Annex VI).

Q18. The IMDG Code became mandatory on

A) 1st of January 2000
B) 1st of January 2002
C) 1st of January 2004
D) 1st of January 2006

Correct Answer: C (1st of January 2004)

Explanation: The International Maritime Dangerous Goods (IMDG) Code was adopted by the IMO and became mandatory under SOLAS Chapter VII on 1st January 2004 to ensure the safe transport of dangerous goods by sea.

Q19. How should CAS be aligned with other ship's surveys?

A) Alignment with the renewal survey and no inclusion in the drydock survey
B) Alignment with the intermediate or special survey and no inclusion in the drydock survey
C) Independent of the intermediate survey and drydock survey

Correct Answer: B (Alignment with the intermediate or special survey and no inclusion in the drydock survey)

Explanation: The Condition Assessment Scheme (CAS) for tankers is a mandatory survey requirement for certain ships. It is aligned with the ship's enhanced survey program, typically during the intermediate or special renewal survey, to ensure structural integrity.

Q20. Liquids that have an open-cup flash point at or below 80 Deg F (26.7 Deg C) are classified as

A) Non-volatile Liquids
B) Flammable Liquids
C) Volatile Liquids
D) Combustible Liquids

Correct Answer: B (Flammable Liquids)

Explanation: In accordance with safety standards, liquids with an open-cup flash point at or below 80 degreesF (26.7 degreesC) are classified as 'flammable liquids' due to their ability to form explosive gas mixtures at ambient temperatures.

Q21. Which Annex deals with the prevention of pollution by sewage from ships?

A) Annex VI
B) Annex III
C) Annex IV

Correct Answer: C (Annex IV)

Explanation: MARPOL Annex IV provides international requirements for the prevention of pollution by sewage from ships, covering discharge rates, treatment plants, and equipment standards for various ship types.

Q22. What type of pollution control is addressed by Annex II of MARPOL?

A) Pollution by oil
B) Pollution by noxious liquid substances
C) Pollution by garbage

Correct Answer: B (Pollution by noxious liquid substances)

Explanation: MARPOL Annex II specifically addresses the control of pollution by Noxious Liquid Substances (NLS) carried in bulk, establishing categorization systems (X, Y, Z, OS) for chemicals to mitigate environmental damage.

Q23. What was the last date by which use of hydrochloroflourocarbons were prohibited?

A) 2015
B) 2010
C) 1 Jan 2020

Correct Answer: C (1 Jan 2020)

Explanation: Under MARPOL Annex VI regulations concerning ozone-depleting substances, the use and installation of equipment containing hydrochlorofluorocarbons (HCFCs) were strictly prohibited globally from 1st January 2020.

Q24. When did Annex VI of Marpol enter into force?

A) 1 July 1992
B) 19 May 2006
C) 31 December 1988

Correct Answer: B (19 May 2006)

Explanation: MARPOL Annex VI, which deals with the prevention of air pollution from ships by regulating sulfur oxides, nitrogen oxides, and ozone-depleting substances, officially entered into force on 19 May 2006.

Q25. What is the open cup flash point of the petroleum liquids that are classified as combustible liquids?

A) Above 80 Deg F (26.7 Deg C)
B) Above 60 Deg F (15.6 Deg C)
C) Below 80 Deg F (26.7 Deg C)
D) Below 60 Deg F (15.6 Deg C)

Correct Answer: A (Above 80 Deg F (26.7 Deg C))

Explanation: Combustible liquids are those with a flash point higher than 80 degreesF (26.7 degreesC). Because they do not readily vaporize at ambient conditions, they are considered less immediately hazardous than flammable liquids but still require safety precautions.

Q26. ____ is a lowest temperature at which a substance catches fire.

A) Critical temperature
B) Ignition temperature
C) Eutectic temperature

Correct Answer: B (Ignition temperature)

Explanation: Ignition temperature is the minimum temperature to which a substance must be heated so that it will self-sustain combustion without the need for an external spark or flame.

Q27. What is an inert condition?

A) When the oxygen has been eliminated from a tank
B) When the oxygen is 10% by volume in a tank
C) A condition in which the oxygen content throughout the atmosphere of a tank has been reduced to 8% or less by volume by the addition of inert gas.
D) When the oxygen is 21% by volume in a tank

Correct Answer: C (A condition in which the oxygen content throughout the atmosphere of a tank has been reduced t

Explanation: An inert condition is defined as an atmosphere where the oxygen content is reduced to 8% or less by volume, usually by injecting flue gas or nitrogen, making the tank atmosphere non-combustible.

Q28. What factor mainly affects Vapour cloud dilution rate?

A) Quantity of Spill
B) Weather conditions
C) Thermal inversion
D) Diameter of the vapour cloud

Correct Answer: B (Weather conditions)

Explanation: Weather conditions, specifically wind speed and direction, are the primary drivers for vapor cloud dispersion and dilution. They dictate how quickly a released gas plume is mixed with fresh air, reducing its concentration below the LFL.

Q29. The substance involved in combustion is called

A) Combustible
B) Non-combustible
C) Inflammable substances

Correct Answer: A (Combustible)

Explanation: The substance that undergoes the chemical reaction of oxidation during fire is technically referred to as the combustible material or fuel.

Q30. What is cold work?

A) Carrying out work in a cold atmosphere

B) Work that cannot create a source of ignition

C) Carrying out work on reefer containers

D) Work that can be done only in cold temperatures

Correct Answer: B (Work that cannot create a source of ignition)

Explanation: Cold work refers to maintenance activities that do not involve heat, sparks, or potential sources of ignition, such as painting or manual cleaning, thereby not requiring a 'Hot Work' permit.